CPT202 Software Engineering Group Project

Assignment 3 – Project Individual Report

Start date	28th March 2026
Submission date	31st May 2026
Assignment type	Individual assignment
Percentage in final marks	50%
Late submission policy	5% of the awarded marks shall be deducted for each working day after the submission date, up to a maximum of five working days. Late submission for more than five working days will not be accepted.
Important notes	• Plagiarism results in award of ZERO mark; all submissions will be checked by Turnitin. • The formal procedure for submitting coursework at XJTLU is strictly followed. Submission link on Learning Mall will be provided in due course. The submission timestamp on Learning Mall will be used to check late submission. • **Generative AI Policy:** The use of Generative AI for content generation is **not permitted**.

Learning Outcome

- A. Work effectively as part of a development team, demonstrating communication and interpersonal skills to collaboratively design and develop a software system.
- B. Explain the software development process, including methods and challenges in deploying systems to meet business goals.
- C. Specify and analyze the requirements of a software system using appropriate techniques.
- D. Produce and evaluate documentation that supports the software development lifecycle.
- E. Analyze legal, social, ethical, and professional issues involved in software development and deployment.

General instruction

In this assignment, you are required to write an individual report describing your contribution to the final solution of the group project. The report should focus on your own work and contribution. However, where relevant, your work must be explained in the context of the team's development process (e.g., sprint planning, task allocation, integration, testing, and change handling).

The report must follow the format specified in the Individual Report Template provided in the Assignment Package. Please note that the report must be within a 4,000-word limit, excluding the following: (1) the cover page containing the student ID, (2) the Table of Contents page, (3) references, and (4) appendices. Appendices may be used to provide supporting evidence (e.g., PBIs, UML diagrams, ERD, screenshots, commit logs, test cases/results, and selected code excerpts). Evidence included in the appendices must be clearly referenced and interpreted in the main report; appendices alone do not constitute analysis.

Your report must demonstrate your understanding of the software development process through evidence drawn from your own project work. Descriptive statements without project-specific evidence will receive limited marks. Appropriate evidence may include backlog items, sprint board snapshots, commit history, pull requests, testing records, deployment logs, or screenshots of issue tracking and team collaboration artefacts.

If Scrum terminology is used (e.g., sprint, backlog, PBI, review, retrospective), the report should reflect how these were actually applied in your project, rather than merely providing textbook definitions.

The marking criteria and weighting are based on the following factors:

1. Report quality (15%):
   The report should be well presented, with language that is clear, concise, and free from ambiguity. Where appropriate, visual aids such as images, diagrams, and charts should be used to improve readability and support understanding. All visual elements must be correctly labelled and explicitly referenced in the report text.

   All visuals and appendix materials used as evidence must be clearly referenced and interpreted in the main report. Unreferenced appendix materials will not be credited.

2. Software development process (20%):
   You should evaluate and discuss how software development was carried out in practice in your project, with particular focus on your own contribution within the team process, supported by clear evidence.

   If Scrum terminology and/or practices are used, the report should explain how they were applied in your project, rather than only describing concepts. Project-specific examples and evidence should be used where possible (e.g., PBIs, sprint/iteration tasks, task tracking records, commits/pull requests, and testing/integration activities).

3. Software Design (25%):
   The design and implementation of all PBIs that you individually developed should be presented at a general level, with clear justification for the design and implementation choices made. UML diagrams should be used, where appropriate, to support your explanation.

   Please focus your detailed discussion on two essential PBIs to which you have made the main and evidenced contribution. For each PBI, discuss the implementation from the following aspects:
   - **Logical flow:** You may present and discuss an activity diagram and/or sequence diagram.
   - **Database:** Discuss how the database design supports the PBI.

- **Object states:** Discuss the state changes of the objects involved and how these state changes affect object behavior
- **UI:** Discuss how the UI design supports delivery of the PBI.

In addition to presenting the final design and implementation of your PBIs, you should explain the key design decisions and trade-offs made during development (including changes from earlier versions, where applicable), and how these decisions were validated through testing and/or feedback.

4. Change Management (15%):
   Discuss how requirement changes were handled (or should be handled) in the context of your project. You should use at least one specific project example. If no actual change occurred in your assigned work, you may use a realistic change scenario based on the project context.

   Your discussion should explain the impact of the change on backlog/PBIs, design, implementation, testing, and team coordination.

5. Legal, social, ethical, and professional issues (15%)
   Identify and discuss any legal, social, ethical, and professional issues that influenced the project, and propose possible solutions to these issues. The discussion should be grounded in the project you worked on (e.g., using specific examples from your own project).

   If the parts you personally worked on did not directly involve such issues, you should address this section from the perspective of the overall project.

   The discussion should not only identify relevant issues, but also explain how they influenced development decisions, documentation, testing, or deployment practices within the project.

6. Conclusion (10%)
   The conclusion should include specific process-related lessons learned (e.g., planning, communication, estimation, integration, testing, documentation, and change handling), as well as practical improvements for future iterations.

You must submit the report in PDF format via Learning Mall and name the report as CPT202-A3-Name-StudentID.pdf.